Where are the Crimes?

STATS 765 - Milestone 2

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1 Goal

The main goal of this project is to see what features in census Statistical Area 2 (areas of 1,000 to 4,000 people) are the highest predictors of crime. My specific section is focusing on the main means of transport to work and the ethnic makeup of the Statistical Area.

2 Data Source

The data for this analysis is coming from 3 main sources:

Dataset	Detail	Dimensions
Census	Method of Transport to and from SA2	65056×15
Meshblock	Mapping table for meshblock to SA2 lookup	57539×55
Victimisations	Victimisation data with meshblock and timestamp	1133581×17

The ANZSOC_Full_data is from the NZ police, which has the place and month the reported victimization took place, and a meshblock to the SA2 table to join the police data. data with the census data, and for this write-up, I am looking at the main means of travel to work and the population change by ethnic group from the 2023 New Zealand census.

3 Data Loading

3.1 Missing/Redacted Data

Using the documentation from StatsNZ, any -999 is listed as a redacted value due to confidentiality. Since we do not have extra information to input these values, we can treat them as `NA`. In the Victimisation data, there are 7497 meshblocks and area units that are redacted, so we cannot use those as well.

3.2 Date Range

The Victimisation data covers the range of 8/2021 to 7/2025, while the census has values for 2018 and 2023. Therefore, to get the most accurate predictions from the data, we are only going to look at data from 2023. For the Victimisation data the requires us to drop all columns that did not occure in 2023. Given this reduction in data size and increasing the likelihood of some Regions having none to a few data points, we also decided to focus on the 4 major cities of Auckland. This brought the victimization data to 173203 observations. The census data is structured differently, with each row being a start and end point on a commute, and the year values are stored in columns with the year and transport type. In order to focus on the 2023 data, we dropped all the dated columns that were not for 2023.

3.3 Data Dimensionality

Unlike most of the census data, the Transport data only has 15 features but has multiple entries for each SA2, with each row being the SA2 the individual is commuting from and the SA2 they are commuting to. Therefore, to get the best picture of how crime might be affected by commuting, we summed the total number of individuals commuting from a location, as well as the total commuting to a location. This way, there is now data on the total number of people leaving and entering each SA2. But looking at the data, we have 65056 pairs of Residency and Workplace with no missing data. But the interesting question this introduces is how the difference between residence location and work location relates to crime. This brings us to 2280 observations with 24 features. The Meshblock data is just for combining Meshblocks with the SA2, so we can drop all but the 2 relevant columns and rename them for ease of use. With the Victimization data, we are most interested in the amount of victimization, the location, and the crime type, so all the other information was dropped to get us to 9 features. We can then join all the tables together to get the final dataset to analyze.

Dataset	Original Size	Final Size
Victimisation	1133581×17	173203×9
Meshblock	57539×55	$\$57539 \times 2$
Means of Transportation	65056×15	2280×24

4 Data Exploration

4.1 Distribution of Commited Offenses

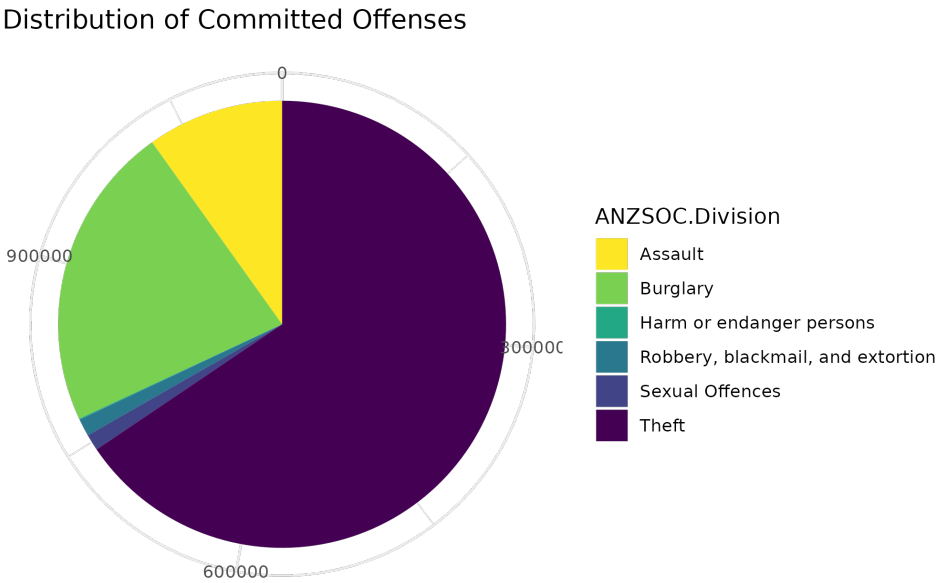


Figure 1: Distribution of Committed Offenses

The majority of crimes are Theft by a large margin, while Sexual Offences and Robbery and blackmail is the least common type of crime.

4.2 Criminal Activity Hotspots

Statistical Area 2 Name	Number of Victimizations
Palmerston North Central	1939
Rotorua Central	1791
Hamilton Central	1612
Te Rapa South	1560
Hutt Central North	1342
Napier Central	1240
Sylvia Park	1138
Whanganui Central	1116
Tauranga Central	1104
Hastings Central	1072

4.3 Commuter Distributions

The first thing to look at is the distribution of commuters.

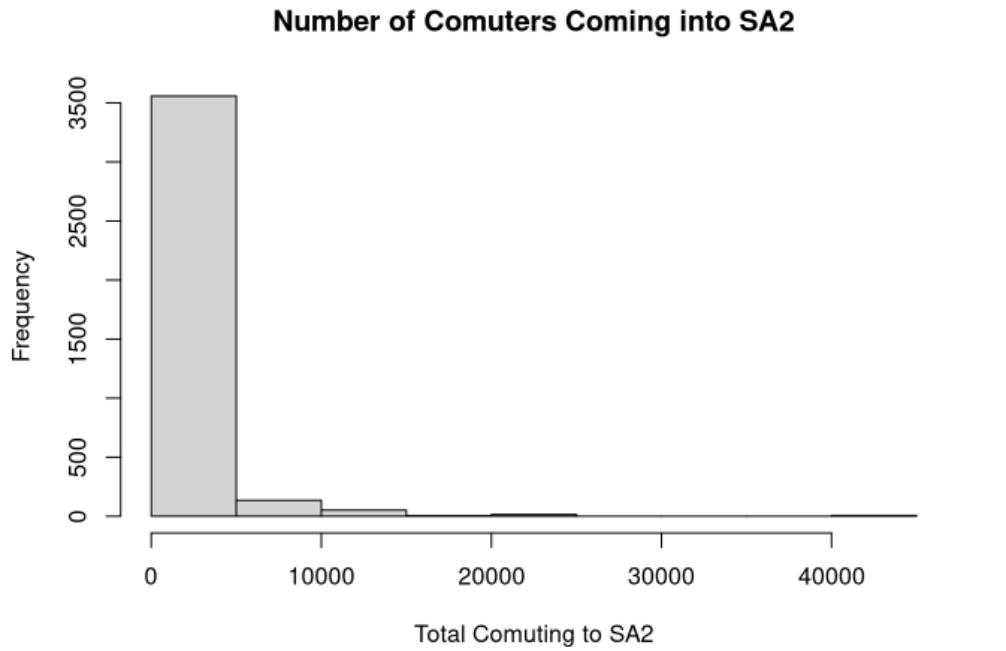


Figure 2: Distribution of Commuters Into

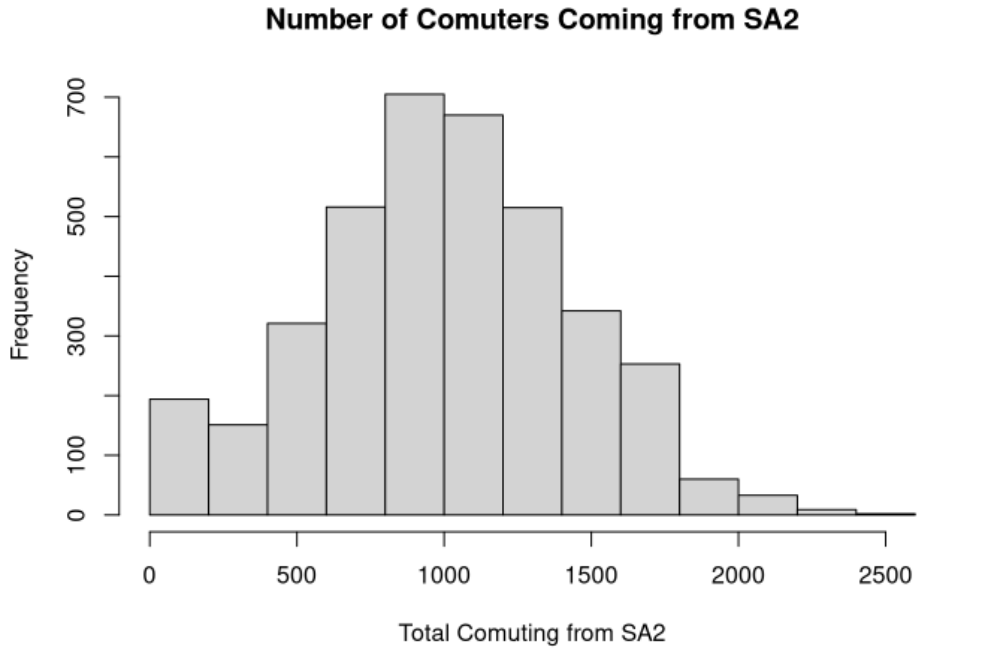


Figure 3: Distribution of Commuters From

These indicate that there is a normal spread of locations people are commuting leaving, with most individuals leaving from an SA2 with about 1000 other commuters. But the people commuting in are spreading out to a large number of SA2s with a smaller number of people.

4.4 SA2 Commuters Coming from versus Coming Out

Now we need to check if there is any difference in crime between SA2 with more people commuting from versus SA2 with more people commuting in.

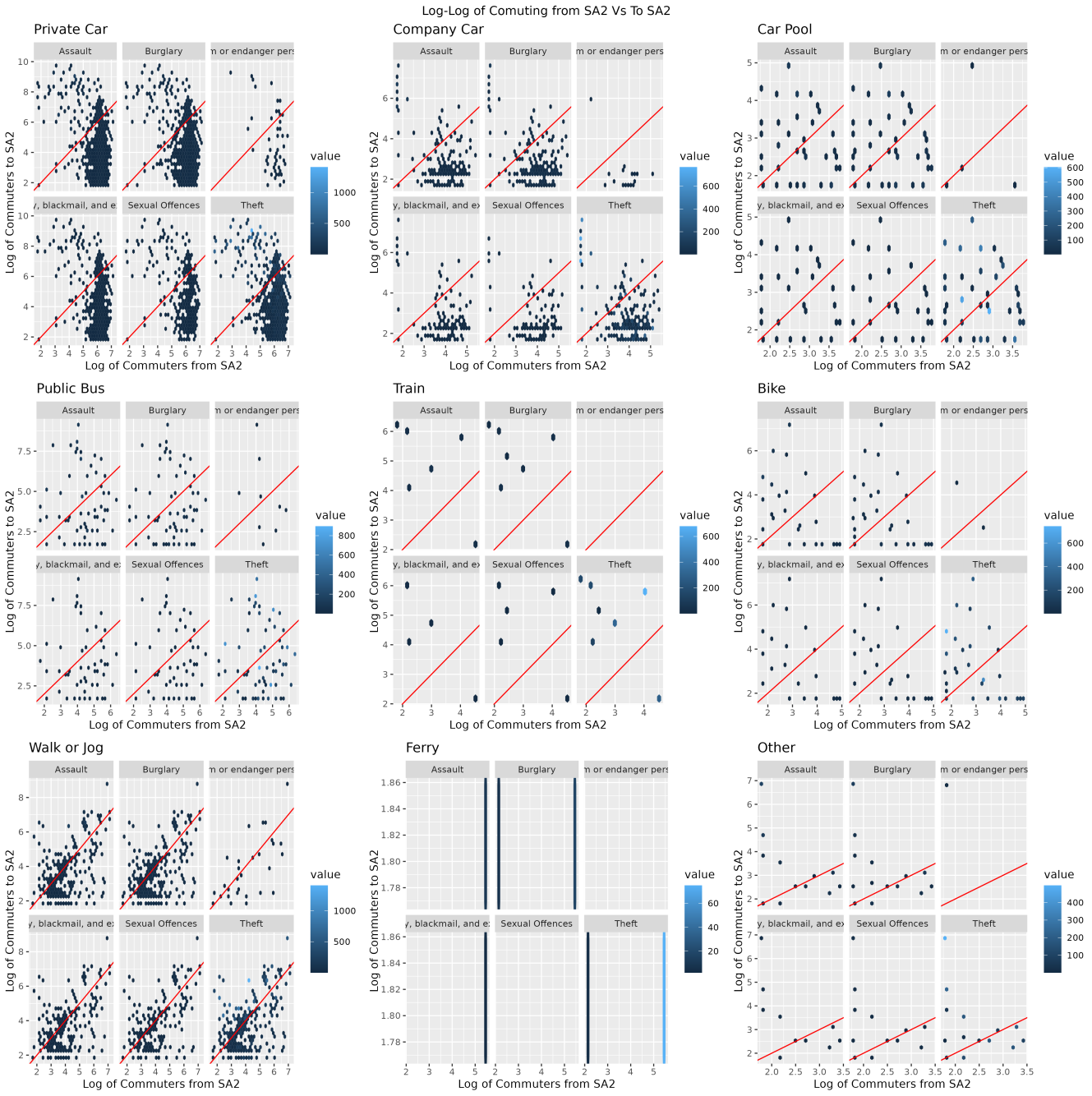


Figure 4: Log-Log Commuting From SA2 Vs to SA2

The above chart is a breakdown of the victimization by ANZSOC Division and method of travel. The points on the graph are the number of commuters leaving the SA2 on X and the number of commuters entering the SA2 on Y. Points below the red line have more commuters leaving than entering, while points above have more commuters

entering. The values of the hexes are the mean of all the binned values.

While being a very complicated graph, it does show a few interesting points. There is a large cluster of victimizations for SA2, where a large number of people live in their cars. This could be because people leave their house in their car, and a few individuals are coming into that area, which might make the area feel emptier, which could encourage more types of crime.

Due to the distribution of most people going towards SA2 with fewer people commuting into it, there is a bit of a bias in the graph, with each SA2 being more likely to have people leaving than entering it, and we are not seeing a clear divide in crime being more likely in more residential areas versus commercial areas. The main bit of information is the higher the population the more crime.

4.5 Victimization Rate Versus Crime Rate

Since there was no clear divide in areas where people are mostly coming or leaving, let's see if there is a trend in the rate of commute type versus victimization.

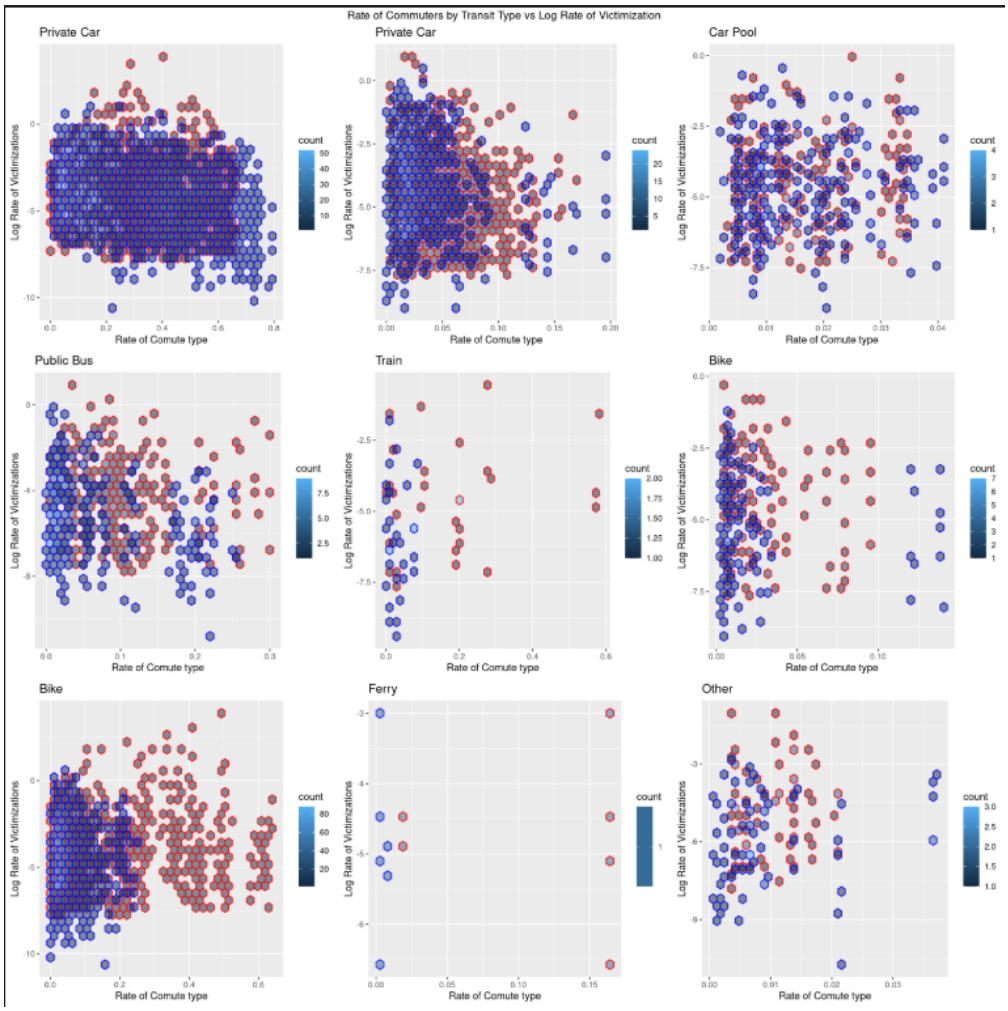


Figure 5: Log-Log Commuting From SA2 Vs to SA2

The blue hexes are the SA2s people are commuting into, while the red is where people are commuting from.

These plots show that there is almost no correlation between the rate of commute type and the crime rate.

5 Analytical Plan

The next step would be establishing relationships between all predictors against a normalized count of victimizations (/ 1000 Households). There is a strong potential for income and crowding to explain the number of victimizations, hence the possibility (polynomial) of a non-linear relationship would be explored. All the socio-economic factors, such as vehicle ownership and access to telecommunication, should be examined in correlation with income and crowding. Establish a geographic context by comparing the patterns in the city center, suburban, and rural settings. Observe the outcome of the backward model selection based on these factors. Build a regression model to predict potential high-risk areas based on predictors. Since most explanatory variables here are numeric, XGBoost may be used. Observe the effects of excluding outliers. Identify and implement ways for effective visual geographical reporting.

6 Appendix

```
library(Hmisc, include.only = c('describe'))
library("tidyverse")
library(grid)
library(gridExtra)
library(hexbin)
library(igraph)
```

```
crime_data = read.csv("data/ANZSOC_Full Data_data.csv") |>
  mutate(Year.Month = my(Year.Month)) |>
  mutate(Month = month(Year.Month), Year = year(Year.Month))|>
  select(ANZSOC.Division, Victimisations, ANZSOC.Group, Meshblock,
         Occurrence.Hour.Of.Day,Territorial.Authority,Month, Year, Year.Month)
describe(crime_data)

focused_crime = crime_data |> # limmiting data to the 4 major NZ cities
filter(Year == 2023 & Territorial.Authority
%in% c("Auckland.", "Wellington City.", "Hamilton City.", "Christchurch City."))
describe(focused_crime)
```

```
crime_data |>
    group_by(ANZSOC.Division) |>
    dplyr::summarize(count_by_subdivision = sum(Victimisations,
    na.rm = T),
    .groups = "drop") |>
    ggplot(aes(x = "",
               y = count_by_subdivision,
               fill = `ANZSOC.Division`)) +
    geom_bar(stat = "identity", width = 1) +
    scale_fill_viridis_d(direction = -1) +
    coord_polar("y", start = 0) +
    labs(
        title = "Distribution of Committed Offenses",
        x = "",
        y = ""
    ) +
    theme_minimal()
```

```
missing_loc = crime_data |> filter(Meshblock < 0) |> nrow()
cat("missing location data: ", missing_loc,'\n')
```

```
names = c("MB2023_V1_00" = "Meshblock", "SA22023_V1_00" = "SA2")
meshblock_data = read.csv("data/meshblock-higher-geographies-2023-generalised.csv") |>
  select(MB2023_V1_00, SA22023_V1_00) |>
  dply::rename(replace = names)

describe(meshblock_data)
```

```
names = c("SA22023_V1_00_usual_residence_address" = "SA2",
            "SA22023_V1_00_NAME_ASCII_usual_residence_address" = "Residence.Name",
            "SA22023_V1_00_workplace_address" = "SA2.Workplace",
            "SA22023_V1_00_NAME_ASCII_workplace_address" = "Workplace.Name")
work_travel = read.csv(
  "data/2023-census-main-means-of-travel-to-work-by-statistical-area-2.csv") |>
  rename(replace = names) |>
  select(-any_of(c("SA22023_V1_00_NAME_workplace_address",
```

```
  "SA22023_V1_00_NAME_usual_residence_address"))) |>
  select(!contains("2018")) |>
  mutate_if(is.numeric, ~ifelse(.x == -999, 0, .x))

  describe(work_travel)

from = work_travel |>
  select(!c(SA2.Workplace, Workplace.Name)) |>
  group_by(SA2, Residence.Name) |>
  summarise_if(is.numeric, sum)

to = work_travel |>
  select(!c(SA2, Residence.Name)) |>
  group_by(SA2.Workplace, Workplace.Name) |>
  summarise_if(is.numeric, sum)

from_to = from |>
  left_join(
    to |> select(!Workplace.Name),
    by = c("SA2" = "SA2.Workplace"),
    suffix = c("_from",  "_to")
  )

describe(from_to)
```

```
crime_SA2 = inner_join(
                      focused_crime,
                      meshblock_data,
                      by = "Meshblock")

total_crime_sa2 = crime_SA2 |>
  group_by(SA2, Territorial.Authority, ANZSOC.Division) |>
  summarise_at("Victimisations", sum) |>
  inner_join(from_to, by = "SA2")
```

```
hist(total_crime_sa2$X2023_Total_stated_to,
main="Number of Comuters Coming into SA2",
xlab= "Total Comuting to SA2")
```

```
hist(total_crime_sa2$X2023_Total_stated_from,
main="Number of Comuters Coming from SA2",
xlab= "Total Comuting from SA2")
```

```
p1 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
                    aes(
                        x=log(X2023_Drive_a_private_car_truck_or_van_from),
                        y = log(X2023_Drive_a_private_car_truck_or_van_to),
                        z = Victimisations),
                        fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Private Car",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
```

```
p2 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
              aes(
                  x=log(X2023_Drive_a_company_car_truck_or_van_from),
                  y = log(X2023_Drive_a_company_car_truck_or_van_to),
                  z = Victimisations),
                  fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Company Car",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
p3 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
              aes(
                  x=log(X2023_Passenger_in_a_car_truck_van_or_company_bus_from),
                  y = log(X2023_Passenger_in_a_car_truck_van_or_company_bus_to),
                  z = Victimisations),
                  fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Car Pool",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
p4 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
            aes(
                x=log(X2023_Public_bus_from),
                y = log(X2023_Public_bus_to),
                z = Victimisations),
                fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Public Bus",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
p5 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
              aes(
                  x=log(X2023_Train_from),
                  y = log(X2023_Train_to),
                  z = Victimisations),
                  fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Train",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
```

```
p6 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
              aes(
                  x=log(X2023_Bicycle_from),
                  y = log(X2023_Bicycle_to),
                  z = Victimisations),
                  fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Bike",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
p7 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
              aes(
                  x=log(X2023_Walk_or_jog_from),
                  y = log(X2023_Walk_or_jog_to),
                  z = Victimisations),
                  fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Walk or Jog",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
p8 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
              aes(
                  x=log(X2023_Ferry_from),
                  y = log(X2023_Ferry_to),
                  z = Victimisations),
                  fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Ferry",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
p9 = ggplot(data = total_crime_sa2) +
  stat_summary_hex(
              aes(
                  x=log(X2023_Other_from),
                  y = log(X2023_Other_to),
                  z = Victimisations),
                  fun = function(x) mean(x)) +
  geom_abline(color = 'red') +
  facet_wrap(~ANZSOC.Division) +
  labs(
    title = "Other",
    x = "Log of Commuters from SA2",
    y = "Log of Commuters to SA2"
  )
```

```
gt1 = ggplotGrob(p1)
gt2 = ggplotGrob(p2)
gt3 = ggplotGrob(p3)
gt4 = ggplotGrob(p4)
gt5 = ggplotGrob(p5)
gt6 = ggplotGrob(p6)
gt7 = ggplotGrob(p7)
gt8 = ggplotGrob(p8)
gt9 = ggplotGrob(p9)

grid_plot = grid.arrange(gt1, gt2, gt3, gt4, gt5,
                         gt6, gt7 ,gt8, gt9,
                         nrow=3,
                         top = "Log-Log of Comuting from SA2 Vs To SA2")
```

```
# find the rate of transport use vs crime
p1 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Drive_a_private_car_truck_or_van_from > 0) |>
  filter(X2023_Drive_a_private_car_truck_or_van_to > 0) |>
  mutate(rate_from_private_vehical =
  X2023_Drive_a_private_car_truck_or_van_from / X2023_Total_stated_from) |>
  mutate(rate_to_private_vehical =
  X2023_Drive_a_private_car_truck_or_van_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from =
  (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to =
  (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
  geom_hex(aes(x=rate_from_private_vehical,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to_private_vehical,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5) +
  labs(
    title = "Private Car",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p2 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Drive_a_company_car_truck_or_van_from > 0) |>
  filter(X2023_Drive_a_company_car_truck_or_van_to > 0) |>
  mutate(rate_from =
  X2023_Drive_a_company_car_truck_or_van_from / X2023_Total_stated_from) |>
  mutate(rate_to =
  X2023_Drive_a_company_car_truck_or_van_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from =
  (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to =
  (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
```

```
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
  labs(
    title = "Private Car",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p3 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Passenger_in_a_car_truck_van_or_company_bus_from > 0) |>
  filter(X2023_Passenger_in_a_car_truck_van_or_company_bus_to > 0) |>
  mutate(rate_from =
  X2023_Passenger_in_a_car_truck_van_or_company_bus_from / X2023_Total_stated_from) |>
  mutate(rate_to =
  X2023_Passenger_in_a_car_truck_van_or_company_bus_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from =
  (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to =
  (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
  labs(
    title = "Car Pool",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p4 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Public_bus_from > 0) |>
  filter(X2023_Public_bus_to > 0) |>
  mutate(rate_from = X2023_Public_bus_from / X2023_Total_stated_from) |>
  mutate(rate_to = X2023_Public_bus_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from = (
    Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to =
  (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
```

```
  labs(
    title = "Public Bus",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p5 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Train_from > 0) |>
  filter(X2023_Train_to > 0) |>
  mutate(rate_from = X2023_Train_to / X2023_Total_stated_from) |>
  mutate(rate_to = X2023_Train_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from =
  (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to = (
    Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
  labs(
    title = "Train",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p6 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Bicycle_from > 0) |>
  filter(X2023_Bicycle_to > 0) |>
  mutate(rate_from = X2023_Bicycle_from / X2023_Total_stated_from) |>
  mutate(rate_to = X2023_Bicycle_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from = (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to = (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
  labs(
    title = "Bike",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p7 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Walk_or_jog_from > 0) |>
```

```
  filter(X2023_Walk_or_jog_to > 0) |>
  mutate(rate_from = X2023_Walk_or_jog_from / X2023_Total_stated_from) |>
  mutate(rate_to = X2023_Walk_or_jog_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from =
  (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to =
  (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
  labs(
    title = "Bike",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p8 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Ferry_from> 0) |>
  filter(X2023_Ferry_to > 0) |>
  mutate(rate_from = X2023_Ferry_from / X2023_Total_stated_from) |>
  mutate(rate_to = X2023_Ferry_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from =
  (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to =
  (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
  labs(
    title = "Ferry",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

p9 = total_crime_sa2 |>
  filter(X2023_Total_stated_from > 0) |>
  filter(X2023_Total_stated_to > 0) |>
  filter(X2023_Other_from> 0) |>
  filter(X2023_Other_to > 0) |>
  mutate(rate_from = X2023_Other_from / X2023_Total_stated_from) |>
  mutate(rate_to = X2023_Other_to / X2023_Total_stated_to) |>
  mutate(Victimisations_rate_from =
  (Victimisations / X2023_Total_stated_from)) |>
  mutate(Victimisations_rate_to =
  (Victimisations / X2023_Total_stated_to)) |>
  ggplot() +
```

```
  geom_hex(aes(x=rate_from,
               y=log(Victimisations_rate_from)),
           color = 'red', alpha=.5) +
  geom_hex(aes(x=rate_to,
               y=log(Victimisations_rate_to)),
           color = 'blue', alpha = .5)+
  labs(
    title = "Other",
    x = "Rate of Comute type",
    y = "Log Rate of Victimizations"
  )

gt1 = ggplotGrob(p1)
gt2 = ggplotGrob(p2)
gt3 = ggplotGrob(p3)
gt4 = ggplotGrob(p4)
gt5 = ggplotGrob(p5)
gt6 = ggplotGrob(p6)
gt7 = ggplotGrob(p7)
gt8 = ggplotGrob(p8)
gt9 = ggplotGrob(p9)

grid.arrange(gt1, gt2, gt3, gt4, gt5,
             gt6, gt7, gt8, gt9,
             nrow=3,
             top = "Rate of Commuters by Transit Type vs Log Rate of Victimization "
             )
```